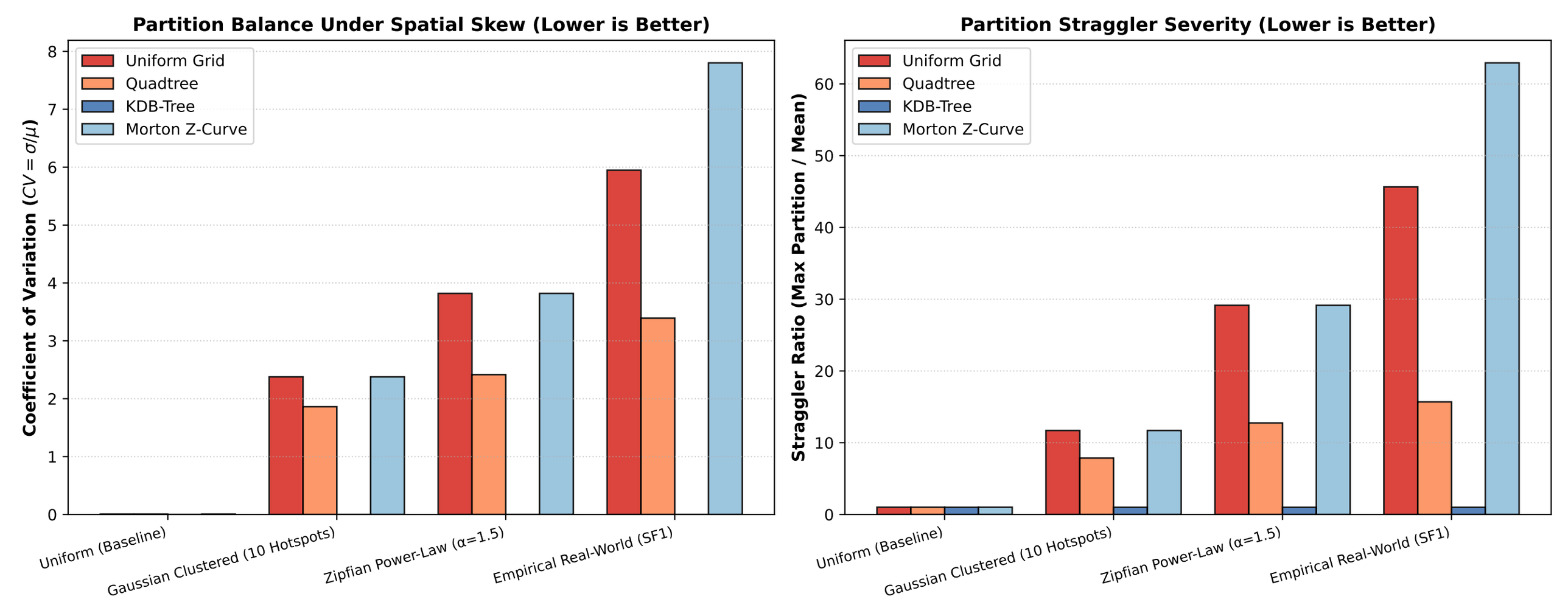
Partition Balance Under Spatial Skew (Lower is Better)
Coefficient of Variation (CV = σ/μ)
Uniform Grid
Quadtree
KDB-Tree
Morton Z-Curve
0
1
2
3
4
5
6
7
8
Uniform (Baseline)
Gaussian Clustered (10 Hotspots)
Zipfian Power-Law (α=1.5)
Empirical Real-World (SF1)
Partition Straggler Severity (Lower is Better)
Straggler Ratio (Max Partition / Mean)
Uniform Grid
Quadtree
KDB-Tree
Morton Z-Curve
0
10
20
30
40
50
60
Uniform (Baseline)
Gaussian Clustered (10 Hotspots)
Zipfian Power-Law (α=1.5)
Empirical Real-World (SF1)